

ALIAHUNIVERSITY

End Semester Examination (Spring Semester)2023

(B. Tech 3rd Year 6th Semester)

SubjectName: Software Engineering

FullMarks: 80

SubjectCode: CSEUGPC17

Time: 3 Hours

Group A

Answer all 10 questions

10x1=10

1. Fill in the blanks with one word

- Full form of COCOMO is _____.
- A GUI interface is a type of _____ software.
- Full form of PERT is _____.
- Beta test is performed by _____ users.
- A design solution is said to be _____ modular, if the different modules in the solution have high cohesion and their inter- module couplings are low.
- _____ between two modules is a measure of degree of interaction between the two modules.
- _____ is a measure of the functional strength of a module.
- Empirical estimation, Heuristic techniques and _____ estimation techniques are the three main project estimation techniques.
- The formula for computation of Cyclomatic complexity of a program from an inspection of CFG is _____.
- The formula to calculate the Slack Time is _____.

Group- B Answer any 6 questions 6x5=30

- ✓2. Explain why spiral model is also called a Meta model.
- ✓3. Describe the limitations of Classical Waterfall Model and how do we overcome it using Iterative model.
- ✓4. Explain Chief Programmer Team, Democratic Team and Mixed Control Team Organisation.
- ✓5. What are the two main approaches to design black box test cases?
- ✓6. What is the type of projects in which you will use prototyping model, evolutionary model and spiral model?
- ✓7. Describe Iterative Waterfall Model with a diagram.
8. Suppose the estimated development time and cost using Putnam's expression has come out to be 1 Year and ₹ 100,000 respectively. What will be the new cost if we have to develop it within 3 months?

Group- B Answer any 4 questions 4x10=40

- ✓9. Write a short code to find whether a number is odd or even. Draw a CFG and hence find the cyclomatic complexity. 2+8=10
- ✓10. Using black box testing approach find the probable test cases for a software that computes the square root of an input integer which can assume values in the range of 0 to 10000.
- ✓11. Explain in details the different classification of cohesiveness. 10
12. Explain the two popular Empirical Estimation Techniques. What is the main shortcoming of Basic and Intermediate COCOMO models? How can it be solved? 6+1+3
- ✓13.
 - a) Describe the shortcomings of LOC as metric for project size estimation.
 - b) Assume that the size of an organic type software product has been estimated to be 60,000 lines of source code. Assume that the average salary of software developers is Rs. 80,000 per month. Determine the effort required to develop the software product, the nominal development time, and the cost to develop the product. 4+6

Aliah University

Odd-Semester (Spring) Examination - 2023

(For 6th Semester BTech. CSE)

Paper Name: Computer Network

Paper Code: CSEUGPC19

Full Marks: 80

Time: 3 Hrs

Group – A

(5X2=10)

1. Answer all –

- Write the full form of TCP & UDP.
- What are the lengths of MAC and IP addresses?
- What is the difference between UTP & STP cable?
- What is Hub and Repeater?
- How do you differentiate Layer 2 & Layer 3 switch?

Group – B

(Answer any 6 questions)

(6X5=30)

- Discuss about the collision and broadcast domain. Draw a schematic diagram for both of them in a sample network scenario.
- Differentiate between guided and unguided media. Why twisted pair cables are twisted?
- What do you mean by connection-oriented and connection-less protocol? Discuss with example.
- White the properties of three network devices namely – Hub, Switch and Router.
- Write and discuss about different duties of Network layer.
- Find class of the following IP address:
(a) 00000001 00001011 00001011 11101111
(b) 11110011 10011011 11111011 00001111
- "Millions of Class A IP addresses are wasted" – justify with example.

Group – C

(Answer any 4 questions)

(10X4=40)

9.

- Compare between TCP/IP and OSI model.
- What is collision in a Network? Discuss about ALOHA protocol with schematic diagram.

[4+6=10]

10.

- What do you mean by flow control and error control? In which layer these are managed?
- Discuss about Stop and Wait ARQ algorithm with schematic diagram.

[3+7=10]

11. a) What is burst error? Give example. How two dimensional parity checking system works? Give example.

- A data word to be sent is 100100; the generator polynomial is $x^3 + x^2 + 1$. Show the actual data to be sent from sender side and check whether any error received at receiver side or not?

[5+5=10]

12. Write short notes on followings (any two)-

[5*2=10]

- Error corrections using hamming code
- Working principle of fiber optics cable
- Classful IP addressing scheme

13.

- Given the address 132.6.17.85, find the network address.
- A router outside the organization receives a packet with destination address 190.240.7.91. Show how it finds the network address to route the packet.
- What is default gateway? How it works?
- Why PING is used? Which protocol is used by PING?

[2+3+3+2=10]

14.

- Explain the principal of ARP and RARP protocol.
- If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?
- What is DHCP protocol?
- Differentiate between packet and frame.

[3+3+2+2=10]

Aliah University
End Semester Examination(Spring Semester) 2023
(For 3rd Year 6th Semester B.Tech(CSE))

Paper Name: Embedded Systems
Paper Code: CSEUGPC21

Full Marks:80
Time:3 hours

Group A

(Answer all questions)

10X1=10

1. What is Embedded System?
2. Differentiate event-triggered and time-triggered system.
3. What is nonoperational quality attribute?
4. What is the use of Programmable Peripheral Interface?
5. Give examples of embedded system used in health care.
6. What is a watchdog timer?
7. What is Real time clock?
8. Give examples of some fourth generation embedded system.
9. Differentiate sensor and actuator.
10. What is oscillator unit?

Group B

(Answer any 5 questions)

6X5=30

1. Differentiate
 - a) I2C and SPI
 - b) Microprocessors and Microcontrollers
2. What are the characteristics of Embedded System?
3. Explain the different components that constitute the core of an embedded system.
4. Explain how Embedded System function in the Data Communication domain.
5. Differentiate Hard and Soft Real Time System.
6. Differentiate Serial and Parallel Communication Interface.

(3+3)

Group C

(Answer any 4 questions)

(4X10=40)

1. a) Explain different classifications of embedded system with examples.
b) Explain the different operational quality attributes in detail.
2. a) Compare the operations of Zigbee and Wi-Fi networks.
b) Discuss about the memory unit used in Embedded Systems.
3. a) Explain Rate Monotonic Scheduling Algorithm.
b) Schedule the following processes by Rate Monotonic Scheduling.

(5+5)

(5+5)

(5+5)

Processes	Execution Time (C)	Time period (T)
P1	3	20
P2	2	5
P3	2	10

4. Write short notes on

(2.5X4)

a) Digital Signal Processor, b) Harvard Architecture, c) FPGA, d) UART

5. a) Define with examples the two types of Communication Interfaces.

b) Explain the sequence of operation for communicating with I2C slave device with the help of a diagram.

(3+7)

Aliah University

Even-Semester (Spring) Examination - 2023

(For 6th Semester BTech. CSE)

Paper Name: Computer Graphics

Paper Code: CSEUGPC20

Full Marks:80

Time: 3 Hrs

(Answer the Questions as per the instruction in each Group and in Sequential Order)

1. **Group – A** Answer all the questions

1x10=10

- What is aspect ratio?
- What is viewport?
- Define frame buffer.
- WCS stands for a) Window Collaboration System b) Window Coordinates System
c) World Coordinates System d) World Computer System
- Light Pen is _____ device.
- What is scaling transformation?
- Which is not an example of emissive display?
a) Plasma panel b) liquid crystal device c) light emitting diode d) thin film electroluminescent display
- On raster system, lines are plotted with
a) Lines b) Dots c) Pixels d) Points
- In Bresenham's line algorithm, if the distances $d_1 < d_2$ then decision parameter P_k is _____.
a) Positive b) Equal c) Negative d) Option a or c
- _____ is a rigid body transformation that moves objects without deformation.

Group – B (Answer any 6 questions)

(6x5=30)

- ✓ 2. Compare Raster and Random Scan Techniques. 5
- ✓ 3. Draw the diagram of CRT and explain its working. 5
- ✓ 4. Prove that two successive rotations are additive. 5
- ✓ 5. What is composite transformation? Explain reflection. 5
- ✓ 6. Explain any two input devices. 5
- ✓ 7. Discuss rotation about a pivot point. 5
8. a) What is line attribute? What are the different attributes of line primitives? 2+3
b) Explain different types of line joins with diagram.
9. Magnify the triangle with vertices A(0,0), B(1,1) and C(5,2) to twice its size while keeping the point C(5,2) fixed. 5
- ✓ 10. What is vanishing point? Define Fractals and Koch Curve? 2+3

Group – C (Answer any 4 questions)

(4x10=40)

11. Explain Bresenham line drawing algorithm with proper mathematical analysis and calculate the pixel positions along a line between A(10,10) and B(18,16) using it. 10
12. a) What are two methods for computing shadows? 5+5
b) Write two types of smooth shading.
- ✓ 13. Explain the Cohen-Sutherland line clipping algorithm with suitable example. 10
- ✓ 14. What is window and viewport? Derive the window to viewport transformation and also identify the geometric transformation involved. 10
- ✓ 15. Explain Sutherland Hodgeman polygon clipping algorithm with suitable example and comment on its shortcoming. 10
- ✓ 16. Write short note on RGB Colour and Grey Color scale. 10
17. Perform 45 degree rotations of triangle A(0,0), B(1,1), C(5,2) about P(-1,-1) 10

(For 6th Semester BTech. CSE)

Full Marks:80

Time: 3 Hrs

- Can decision trees be used for performing clustering?
A. True B. False
- What is the minimum no. of variables/ features required to perform clustering?
A. 0 B. 1 C. 2 D. 3
- Data that sits outside the trend is referred to as a _____.
A. Outlier B. Trend C. Spike D. Both 1 & 2
- What is Big Data?
A. Data with the word 'big' in it B. Data about people who are big
C. Data with a large size D. Data made with a big purpose
- Which of the following things can be accomplished with linear model?
A. Flexibly fit complicated functions B. Uncover complex multivariate relationships
C. Build accurate prediction models D. All of the mentioned
- Which of the following algorithm is most sensitive to outliers?
A. K-means clustering algorithm B. K-medians clustering algorithm
C. K-modes clustering algorithm D. K-medoids clustering algorithm
- Which of the following metrics do we have for finding dissimilarity between two clusters in hierarchical clustering?
1. Single-link 2. Complete-link 3. Average-link
A. 1 and 2, B. 1 and 3 C. 2 and 3 D. 1, 2 and 3
- What is unsupervised learning?
A. features of group explicitly stated B. number of groups may be known
C. neither feature & nor number of groups is known D. none of the mentioned
- Which of the following model has ability to learn?
A. pitts model B. rosenblatt perceptron model C. both rosenblatt and pitts model D. neither rosenblatt nor pitts
- Which of the following statement is False in the case of the KNN Algorithm?
(A) For a very large value of K, points from other classes may be included in the neighborhood.
(B) For the very small value of K, the algorithm is very sensitive to noise.

(C) KNN is used only for classification problem statements.

(D) KNN is a lazy learner.

Group - B

Answer any five question

5 x 6 = 30

1. Describe k-NN classification technique. What do you understand by training and testing set of a classifier?
2. Write down and explain the Hunt's algorithm for decision tree learning.
3. For the following one dimensional data calculate the cost function of k-means after 2 iterations [18, 5, 10, 20, 4, 8, 19].
4. Explain cross validation technique. What is stratified sampling?
5. Explain what will be the minimum and maximum value of GINI index from its definition. Write Bayes classification rule.
6. Write down the algorithm for agglomerative hierarchical clustering. What are the limitations of K-means?
7. Explain the differences between regression and classification. Differentiate supervised, unsupervised and semi-supervised model

Group -C

Answer any four questions

4 x 10 = 40

1. For the one dimensional dataset in question no. 3 in Group-B, perform hierarchical clustering (single linkage) and draw the dendrogram. Write down the advantage of hierarchical clustering. What is 'k' in K-means?
6 + 2 + 2 = 10
2. What do you understand by training and testing set of a classifier? How you measure the performance of a classifier. Give an example/scenario where accuracy measure is misleading.
3 + 4 + 3 = 10
3. Write the Hunt's algorithm for decision tree learning. Given the following data set, which attribute will be chosen first to be split using GINI measure? Data= (feature1, feature2 ,feature3, class: <1, W, 10, P>, <1, X, 20, P>, <2, Y, 10, P>, <3, Y, 10, N>, <1, X, 20, N>)
4 + 6 = 10
4. Explain the basic assumption of Bayes classification. Consider the following data set which explain different conditions that are associated with accidents.:

SNo.	Weather condition	Road condition	Traffic condition	Engine problem	Accident
1	Rain	bad	high	no	yes
2	snow	average	normal	yes	yes
3	clear	bad	light	no	no
4	clear	good	light	yes	yes
5	snow	good	normal	no	no
6	rain	average	light	no	no
7	rain	good	normal	no	no
8	snow	bad	high	no	yes
9	clear	good	high	yes	no
10	clear	bad	high	yes	yes

In this dataset, the target variable accident is a binary categorical variable with yes/no values. There are 4 categorical features: weather condition, road condition, traffic condition, and engine problem. We are interested in building a system which will enable us to decide whether or not road accident occurs. Consider a new data instance X = (Rain, good, normal, no). How would the Naive Bayes classifier classifies X?

2 + 8 = 10

5. Define ROC curve. Explain the significance of the points (0,0), (0,1), (1,0) in ROC space. Assume there are four prediction results from 100 positive and 100 negative instances. The confusion matrices are given as follows: (TP, FN, FP, TN)= (63, 37, 28, 72) for classifier A, (77, 23, 77, 23) for classifier B, (24, 76, 88, 12) for classifier C, (76, 24, 12, 88) for classifier D. Which signify the best prediction result and why? Plot the points in a ROC space.
2 + 2 + 4 + 2 = 10
6. Consider the dataset in question no 3 in Group-B. Assume the label of the seven instances are: [0 1 1 0 1 1 0]. For a sample x=25, find out the label of x using K-NN classification (consider k=3). Differentiate partitional and hierarchical clustering. What signify 'k' in K-NN classification?
6 + 2 + 2 = 10
7. Explain dependent and independent variable in linear regression. What is multivariate regression. For the following dataset find out the best fit line using the linear regression model.

X	2	4	6	8
y	3	7	5	10

3 + 2 + 5 = 10

Aliah University
End-Semester Examination - 2023
(BTech, CSE 6th Semester)

Paper Name: Compiler Design
Paper Code: CSEUGPC18

Full Marks: 80
Time: 3 Hrs

Group - A

(5X2=10)

1. Answer the following questions -

- a) Write name of some cousins of compiler.
- b) How Top-down parsers are different from Bottom-up parsers?
- c) "Any grammar with ambiguity can't be considered as a Parser"-why?
- d) What is left-recursion? How is it removed from a grammar?
- e) Determine the strings generated by this regular expression: $a(bc)^*d$

Group - B

(Answer any 6 questions)

(6X5=30)

- 2. Write a note on various compiler construction tools. Discuss structure of a lex code. 3+2
- 3. What are the regular expressions of valid identifiers, integers and floating constants? Construct Finite Automata to accept valid identifiers, integers and floats. 3+2
- 4. Define Operator grammar. Briefly explain the parsing concept of operator precedence parser. 1+4
- 5. What's the role of Semantic rules in parsing? Differentiate between S -Attributed and L -Attributed Syntax Directed Definitions. Give examples. 1+4
- 6. Calculate LR(1) item sets of the grammar given below - 5

$S \rightarrow aAd \mid bBd \mid aBe \mid bAe$

$A \rightarrow g$

$B \rightarrow g$

Here $N = \{S, A, B\}$ and S is starting symbol

- 7. What is Directed Acyclic Graph (DAG)? How is it different from Abstract Syntax Tree (AST)? Form AST and DAG for following statement: $x = a + (b * c) / d + (b - c) * d + (b * c) \% s$ 1+2+2
- 8. Mention some Peephole optimization techniques with example. Why is it called as Peephole optimization? 4+1

Group - C

(Answer any 4 questions)

(10X4=40)

- 9. a) Calculate FIRST() and FOLLOW() of all the Non-terminals in the following grammar - 3+3

$S \rightarrow ADB \mid DbB \mid Ba$

$A \rightarrow da \mid BD$

$B \rightarrow g \mid \epsilon$

$D \rightarrow h \mid \epsilon$

Here $N = \{S, A, B, D\}$ and S is start symbol

b) What is the second L in LL(k) parsers? Conclude whether the following grammar could be considered as LL(1) parser or not. 1+3

$A \rightarrow rB \mid Ds$

$B \rightarrow AtD$

$D \rightarrow tB \mid \epsilon$

Here $N = \{A, B, D\}$ and A is start symbol

10. a) Mention different roles of Lexical Analyzer. Explain Buffer pair concept, used for Tokenization. 3+4
 b) What is sentinel? How sentinel is better than buffer-pair concept?
 c) Write a Syntax Directed Translation to perform binary to decimal conversion. Consider any grammar suitable for the same. (2+3)+2+3

11. a) What are various ways to optimize a loop? Briefly explain other machine independent code optimization techniques. 3+4
 b) Consider the Syntax Directed Definitions given below. Using the SDTs, what will be the output printed by a bottom-up parser for the input: $2*3+5$? 3

$S \rightarrow ER \ \{\}$

$R \rightarrow *E \ \{\text{printf}("***");\}$

$R \rightarrow \epsilon \ \{\}$

$E \rightarrow F+E \ \{\text{printf}("+");\}$

$E \rightarrow F \ \{\}$

$F \rightarrow (S) \ \{\}$

$F \rightarrow id \ \{\text{printf}("%d", id.lexval);\}$

Here $N = \{S, E, F, R\}$ and S is start symbol

12. a) Find all LR(0) item sets for the following grammar - 4

$S \rightarrow AA$

$A \rightarrow aA$

$A \rightarrow b$

Here $N = \{S, A\}$ and S is start symbol

- b) Construct SLR(1) parsing table and state whether the grammar is SLR(1) or not. 3
 c) Explain full parsing mechanism for an input: **abab**. Is it accepted or not? 3
13. a) Design a 3-Address code for following line of code and represent them into quadruple and triples format. 3+2+2

while(n>0)

{

if(a>5 || b<6)

p++;

else

q++;

n--;

}

- b) Show the Basic blocks in above 3-address code and draw the flow graph to explain flow of control. 3